BIOLOGY 2
0510

# GENERAL CERTIFICATE OF EDUCATION BOARD

General Certificate of Education Examination

**JUNE 2023**

**ORDINARY LEVEL**

| | |
|---|---|
| Subject Title | BIOLOGY |
| Paper No. | 2 |
| Subject Code No. | 0510 |

**Two and half hours**

***Answer FIVE questions, ALL THREE (3) from SECTION A and any TWO (2) from SECTION B.***

***All questions carry 20 marks each. For your guidance, the approximate mark for each part of a question is indicated in brackets.***

***Begin each question on a new page.***

***Give labelled diagrams where these will make the answers clearer.***

***You are advised to read carefully through the question paper before you begin your answers.***

*You are reminded of the necessity for good English and orderly presentation in your answers.*

***In calculations, you are advised to show all the steps in your working, giving your answer at each stage.***


© GCEB2023

## SECTION A

**Answer ALL 3 questions**

**1.** (a) Draw a Spirogyra cell to show its features. **(5 marks)**

(b) Give the functions of any four parts labelled. **(4 marks)**

(c) How is Spirogyra different from an Amoeba? **(6 marks)**

(d) What will happen if a marine amoeba is put in a fresh water environment? **(Total = 20 marks)**

---

**2.** (a) Describe an experiment to measure the rate of water uptake by a leafy shoot. **(9 marks)**

(b) Explain how **three (3)** environmental factors affect the rate of transpiration. **(6 marks)**

(c) Tabulate the differences between a dicotyledonous stem and a monocotyledonous stem? **(Total = 20 marks)**

---

**3.** (a) Define the following terms as used in ecology:

(i) Pollutant.

(ii) Afforestation. **(2+2 marks)**

(b) State the harmful effects of bush burning. **(5 marks)**

(c) Describe the flow of energy in nature. **(6 marks)**

(d) During a bush burning exercise, the quantity of carbon dioxide released was estimated and measured at regular intervals as shown in the table below. **(Total = 20 marks)**

| Time/minutes | 0 | 5 | 15 | 25 | 30 | 35 |
|---|---|---|---|---|---|---|
| $CO_2$/litres | 0 | 10 | 30 | 80 | 70 | 50 |

Draw a graph to represent this information. From the graph, determine the amount of carbon dioxide that will be released during the $10^{th}$ minute. **(5 marks)**

---

## SECTION B

**Answer any 2 questions**

**4.** (a) Draw a neatly labelled diagram to show the position of endocrine glands in a male mammal. **(6 marks)**

(b) State the functions of the mammalian skeleton. **(8 marks)**

(c) Describe what happens when the hand touches a hot object? **(6 marks)**

**(Total = 20 marks)**

---

**5.** (a) Draw a large labelled diagram of the longitudinal section through the mammalian heart, showing the direction of blood flow. **(13 marks)**

(b) Describe the functions of the major components of blood. **(7 marks)**

**(Total = 20 marks)**

---

**6.** (a) Make a large labelled diagram of the male reproductive system of a mammal. **(8 marks)**

(b) What is meant by:

(i) Ovulation,

(ii) Fertilization,

(iii) Pollination? **(2+3+3 marks)**

(c) Bring out **four (4)** differences between an insect pollinated flower and a wind pollinated flower. **(4 marks)**

**(Total = 20 marks)**

---

**7.** (a) What is holozoic nutrition? **(2 marks)**

(b) How is the ileum adapted for absorption of soluble food? **(8 marks)**

(c) Copy and complete the table below.

| Vitamin/mineral | Source | deficiency | function |
|---|---|---|---|
| Vitamin A | Milk, meat etc. | Rickets | Proper vision |
| Iron | | Anaemia | |
| | Oranges, grapes, limes etc. | | Prevents bleeding gums |
| Calcium | | | Helps in muscle contraction |

**(10 marks)**

**(Total = 20 marks)**

---



**GO BACK AND CHECK YOUR WORK**